Robust cascaded PID disturbance observer-based control of an unmanned quadrotor aerial vehicle

Mohammed Hany
Mechatronics Engineering Department
German University in Cairo
mohammed.hany@student.guc.edu.eg

Mohamed Elsayd
Mechatronics Engineering Department
German University in Cairo
mohamed.ramadan@student.guc.edu.eg

Ayman El-Badawy
Mechatronics Engineering Department
German University in Cairo
ayman.elbadawy@guc.edu.eg

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***Abstract*—A disturbance observer (DOB) is incorporated into a cascaded PID control framework for a quadrotor unmanned aerial vehicle (UAV) to improve trajectory tracking under external disturbances. While the cascaded PID structure offers simplicity of implementation, it lacks robustness against time-varying disturbances. To address this, the system is linearized at selected operating points for DOB design, and the robust stability of the Q-filter is verified using the small-gain theorem. Simulation studies with trajectories subject to Dryden wind disturbances demonstrate that the DOB-enhanced controller achieves superior tracking accuracy compared to the conventional cascaded PID controller achieving reductions in position standard deviation of 26.47%, 39.5%, and 59.7% along the X, Y, and Z axes, respectively.**

***Index Terms*—UAV, Quadrotor, cascaded PID, disturbance observer**

I. Introduction

Unmanned Aerial Vehicles (UAVs) have attracted the attention of many researchers for the past two decades. Advancements in micro-controllers, sensors and power storing devices lead to the manufacturing of different types of UAVs [1]. The multi-rotor quadcopter was chosen due to its easier controllability and maneuverability, low price and portability [2]. Quadrotors are widely used in different fields such as disaster management, data collection Internet of Things(IOT) [3], remote sensing, mapping [4], search and rescue, construction and infrastructure inspection, precision agriculture, delivery of goods and real-time monitoring of road traffic [5].

Unfortunately, UAVs represent an unstable coupled system, thus, researchers are interested in designing control systems that enable a UAV to follow a specified trajectory with minimal tracking errors. These controllers include Feedback Linearization Control [6], Sliding Mode Control (SMC) [7], Backstepping control [8], PID control [9], Model Predictive Control (MPC) [10], Linear Quadratic Regulator (LQR) [11], Adaptive control [12] and H_∞ control [13].

Disturbance rejection is one of the most important goals when designing control systems. There are two types of disturbances: external disturbances caused by the environment like wind disturbances and input signal fluctuations [14], and internal disturbances caused by uncertainties like modeling errors, parameter changes and sensor noise [15]. There are many different techniques for disturbance rejection. One of these techniques is using a disturbance observer (DOB). Several papers were written using DOBs in UAV systems. In [16] a time domain DOB-based controller is applied to a quadrotor unmanned aerial vehicle, for [17] a linear dual DOB is used for disturbance rejection for a quadrotor UAV, while in [18] a DOB designed using the H_∞ synthesis techniques is used for disturbance rejection for a hybrid VTOL UAV, for [19] a finite-time disturbance observer combined with a sliding mode controller is used for disturbance rejection for a quadrotor UAV and in [20] a backstepping disturbance observer based controller is designed for trajectory tracking for a multi rotor UAV using two DOBs to estimate force and torque disturbances.

Although DOB-based control has been explored using time-domain and H_∞ techniques, existing literature validates these methods against simplified disturbances (e.g., step or sinusoidal inputs). This work addresses a gap by evaluating the DOB-Cascaded PID integration against the Dryden wind turbulence model on a quadrotor UAV. The robust stability of the chosen filter Q for the disturbance observer is assessed using the small gain theorem. The results are then compared to that of a normal cascaded PID controller.

The paper is organized as follows: section 2 is going to cover the non-linear quadrotor model and the Dryden wind model, section 3 is covering the design of the cascaded PID controller used in the quadrotor system and the linearized model, section 4 is covering the design of the used disturbance observer, section 5 will cover the simulations and results and section 6 is going to cover the conclusion of this paper.

II. Dynamic modeling

A. Modeling the quadrotor dynamics

With the help of [21], the equations of motion were derived for a quadrotor using Lagrange energy technique in the Body-Fixed Axes. For the translational motion, the following equations were derived:

$$\begin{aligned} m\ddot{x} &= T\cos\phi\sin\theta - A_x\dot{x} \\ m\ddot{y} &= -T\sin\phi - A_y\dot{y} \\ m\ddot{z} &= T\cos\phi\cos\theta - mg - A_z\dot{z} \end{aligned} \tag{1}$$

where m is the mass of the quadrotor, g is the acceleration due to gravity, ϕ, θ and ψ are the roll, pitch and yaw angles respectively, A_x, A_y and A_z represent the drag coefficients in the X, Y and Z directions respectively and T represents the

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thrust force which is represented by the following equation:

$$T = k(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \tag{2}$$

where k is the lift factor and ω_i is the angular velocity of the motor. For the rotational motion, the following equations were derived:

$$\begin{aligned} J\frac{d^2\eta}{dt^2} + C\frac{d\eta}{dt} &= B \\ \eta &= \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} \\ B &= \begin{bmatrix} Lk(\omega_4^2 - \omega_2^2) \\ Lk(\omega_3^2 - \omega_1^2) \\ b(\omega_2^2 + \omega_4^2 - \omega_1^2 - \omega_3^2) \end{bmatrix} \end{aligned} \tag{3}$$

where L is the length of the quadrotor's arm, b is the drag factor, while C and J are 3×3 matrices that are shown in the appendix.

B. Dryden wind modeling

In the used model, disturbances are modeled using the Dryden wind model where the gust component of the wind is modeled using the following transfer functions[22]:

$$\begin{aligned} H_u(s) &= \frac{u_d(s)}{\eta_u(s)} = \sigma_u\sqrt{\frac{2V_a}{\pi L_u}}\frac{1}{s + \frac{V_a}{L_u}} \\ H_v(s) &= \frac{v_d(s)}{\eta_v(s)} = \sigma_v\sqrt{\frac{3V_a}{\pi L_v}}\frac{(s + \frac{V_a}{\sqrt{3}L_v})}{(s + \frac{V_a}{L_v})^2} \\ H_w(s) &= \frac{w_d(s)}{\eta_w(s)} = \sigma_w\sqrt{\frac{3V_a}{\pi L_w}}\frac{(s + \frac{V_a}{\sqrt{3}L_w})}{(s + \frac{V_a}{L_w})^2} \\ v_{wx} &= \bar{v}_{wx} + u_d \\ v_{wy} &= \bar{v}_{wy} + v_d \\ v_{wz} &= \bar{v}_{wz} + w_d \end{aligned} \tag{4}$$

where u_d, v_d and w_d represent deviations from the mean velocity, σ_u, σ_v and σ_w are turbulence intensities, L_u, L_v and L_w are the turbulence scale lengths, V_a is the mean wind velocity, η_u, η_v and η_w represent zero-mean white gaussian noise with a unity standard deviation, v_{wx}, v_{wy} and v_{wz} represent the wind velocities across the inertial axes and s is the Laplace variable. The mean wind velocities are chosen to be 0.5 m/s for X and Y and 0.3 m/s for Z.

III. Controller design

The overall system is shown in figure 1. It consists of 5 main parts: the motion profile which generates the reference points, the drone model, the outer loop, the inner loop and the disturbance observer. The cascaded PID controller was used for controlling the quadrotor system due to its simplicity and easier tuning in comparison to other non-linear techniques. The control system consists of two loops: the outer loop and the inner loop.

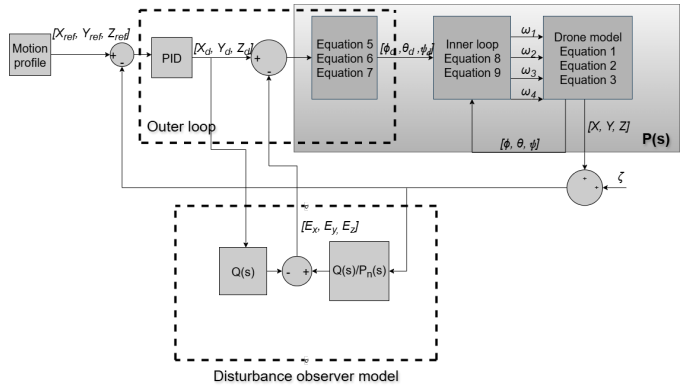

Fig. 1. Detailed overall control scheme with disturbance observer, ζ represents external disturbances

A. Outer loop

The outer loop calculates the difference between the reference position and current position and multiplies each coordinate by its respective PID controller. The resulting values X_d, Y_d and Z_d, represent the desired accelerations in the X, Y and Z directions respectively, are used to calculate the desired pitch angle θ_d, the desired roll angle ϕ_d and the desired thrust T_i using the following equations:

$$\theta_d = \tan^{-1}(\frac{X_d}{Z_d + g}) \tag{5}$$

$$\phi_d = \tan^{-1}(\frac{-Y_d \cos\theta_d}{Z_d + g}) \tag{6}$$

$$T_i = m * \sqrt{X_d^2 + Y_d^2 + (Z_d + g)^2} \tag{7}$$

The desired yaw angle ψ_d is assumed to be 0.

B. Inner loop

In the inner loop the difference between the desired angles and the current angles is calculated and multiplied by PD controllers to generate the desired angular accelerations $\ddot{\phi}$, $\ddot{\theta}$ and $\ddot{\psi}$. The angular acceleration values are used to calculate three torques in the roll, pitch and yaw directions using the following equations:

$$\begin{aligned} \tau_\phi &= I_{xx}\ddot{\phi} \\ \tau_\theta &= I_{yy}\ddot{\theta} \\ \tau_\psi &= I_{zz}\ddot{\psi} \end{aligned} \tag{8}$$

where I_{xx}, I_{yy} and I_{zz} are the products of inertia about the body frame's inertial axes.
The calculated torques in combination with the desired thrust T_i calculated in equation (7) are used to calculate the angular

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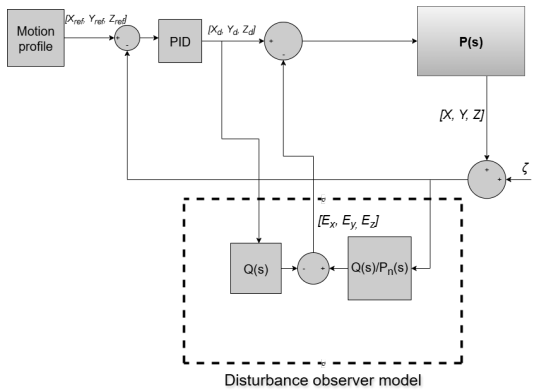

Fig. 2. Simplified model of the overall control scheme

speeds of the four motors using the following equations:

$$\begin{aligned} \omega_1^2 &= \frac{T_i}{4k} - \frac{\tau_\theta}{2kl} - \frac{\tau_\psi}{4b} \\ \omega_2^2 &= \frac{T_i}{4k} - \frac{\tau_\phi}{2kl} + \frac{\tau_\psi}{4b} \\ \omega_3^2 &= \frac{T_i}{4k} + \frac{\tau_\theta}{2kl} - \frac{\tau_\psi}{4b} \\ \omega_4^2 &= \frac{T_i}{4k} + \frac{\tau_\phi}{2kl} + \frac{\tau_\psi}{4b} \end{aligned} \tag{9}$$

C. Linearized model of the quadrotor control system

In order to design the disturbance observer, a linearized model of the gray box in figure 1 which represents the actual plant $P(s)$, which includes the outer loop, the inner loop and the plant, is required to get the necessary transfer functions of the nominal plant $P_n(s)$. Figure 2 shows the simplified quadrotor control system. In order to linearize the quadrotor system, three operating points were chosen corresponding to the UAV motion in each of the three axes with constant velocity. The defined operating points were as follows:

$$\begin{aligned} [\tilde{x}, \tilde{y}, \tilde{z}] &= [x_0, y_0, z_0] \\ [\tilde{\dot{x}}, \tilde{\dot{y}}, \tilde{\dot{z}}] &= [v_x, v_y, v_z] \\ [\tilde{\phi}, \tilde{\theta}, \tilde{\psi}] &= [0, 0, 0] \\ [\tilde{\dot{\phi}}, \tilde{\dot{\theta}}, \tilde{\dot{\psi}}] &= [0, 0, 0] \end{aligned} \tag{10}$$

where for each operating point two of the three values v_x, v_y and v_z are equal to zero while the third value is 5. The linearization was done by the MATLAB Simulink Model Linearizer app. For each operating point, the nominal plant transfer function between one of the three positions (X, Y and Z) and its corresponding desired acceleration (X_d, Y_d and Z_d) that is produced by the PID controller in the outer loop of the control system was computed to be:

$$P_n(s) = \begin{cases} \frac{X(s)}{X_d(s)} & = \frac{0.06901s^3+52.71s^2+3438s+763.8}{s^5+764.3s^4+3846s^3+2600s^2+408s} \\ \frac{Y(s)}{Y_d(s)} & = \frac{0.06901s^3+52.71s^2+3438s+763.8}{s^5+764.3s^4+3846s^3+2600s^2+408s} \\ \frac{Z(s)}{Z_d(s)} & = \frac{1}{s^2+0.53s} \end{cases} \tag{11}$$

IV. Disturbance observer design

The disturbance observer is mainly concerned with predicting the disturbances facing the system and then compensating for it in the controller's output. Figures 1 and 2 show the DOB model where E_x, E_y and E_z represent the estimated disturbances in the X, Y and Z directions respectively, $Q(s)$ represents the used filter and ζ represents the external disturbances. In order to design a disturbance observer two things are needed:

- The inverse of the transfer function of the nominal plant which was already computed in equation (11).
- A filter Q.

The chosen filter Q for the DOB should satisfy the following requirements:

- $Lim_{s->0}Q = 1$ so that at low frequencies $Q(s) \approx 1$ which makes the estimated disturbances $[E_x, E_y, E_z]$ equal to the actual disturbances facing the system.
- The relative degree of Q should be more than or equal to the relative degree of the nominal transfer function as the nominal plant $P_n(s)$ in equation (11) is linear time-invariant system whose relative degree is more than 1, so its inverse $P_n^{-1}(s)$ becomes an improper fraction that needs to be multiplied by a transfer function that satisfies this condition.

A filter that satisfies the previous requirements is able, at low frequencies, to give an accurate estimation of the disturbances facing the system and compensate for it in the controller's output. Consequently, a low pass filter is used and its bandwidth is tuned to be as large as possible to reject disturbances with lower frequencies. However, if the bandwidth is increased, the plant might be affected by noise in the system. To help in designing the used filter, the small gain theorem is used [23] which guarantees the stability of the system with the added filter if the following inequality is satisfied:

$$||Q(s)\Delta(s)||_\infty < 1 \tag{12}$$

which can be rewritten as:

$$||Q(s)||_\infty < \frac{1}{||\Delta(s)||_\infty} \tag{13}$$

where $\Delta(s)$ refers to the uncertainty between the nominal plant and the actual plant. While this study primarily models $\Delta(s)$ as a time delay, in practice, it also encompasses parametric uncertainties such as mismatches in mass, inertia tensors, and aerodynamic drag or thrust coefficients. These parametric errors would increase the magnitude of the uncertainty $||\Delta(s)||_\infty$, at higher frequencies. According to the small-gain theorem (13), the presence of significant parametric mismatch would lower the stability upper bound (the $1/\Delta$ curve), therefore necessitating a more conservative Q-filter with a lower cutoff frequency ω_n to guarantee stability[24].

To evaluate the validity of this assumption, the behavior of the transfer functions of the nominal plant calculated in equation (11) is compared to that of the actual plant when tracking a straight path in each axis.

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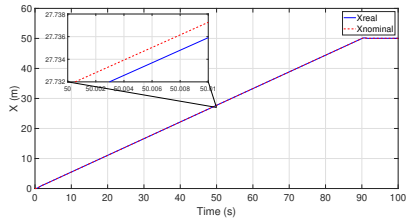

Fig. 3. Comparison of the linearized and real plant responses in the X-axis

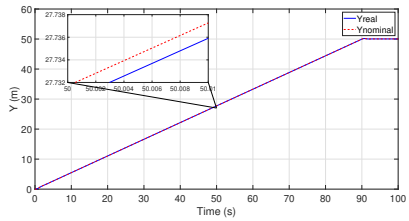

Fig. 4. Comparison of the linearized and real plant responses in the Y-axis.

As seen from figures 3, 4 and 5, the actual plant and the nominal plant behave in a similar manner with only a time delay between them. The time delay between the real plant and the nominal plant in the X,Y and Z directions was found to be 2.5, 2.5 and 1.4 milliseconds respectively.
If the nominal plant is given by the following transfer function:

$$P_n(s) = \frac{B_n(s)}{A_n(s)} \tag{14}$$

Then the actual plant will be given by the following transfer function:

$$P(s) = e^{-sT_d}\frac{B_n(s)}{A_n(s)} \tag{15}$$

where T_d refers to the time delay between the actual plant and the nominal plant.
From equations (14) and (15) we get the following equation:

$$P(s) = e^{-sT_d}P_n(s) \tag{16}$$

which can be rewritten as:

$$P(s) = P_n(s)(1 + \Delta(s)) \tag{17}$$

where:

$$\Delta(s) = e^{-sT_d} - 1 \tag{18}$$

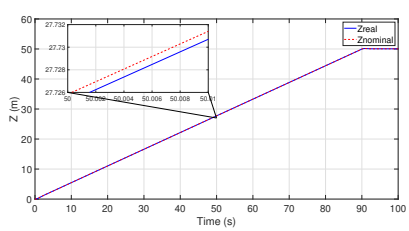

Fig. 5. Comparison of the linearized and real plant responses in the Z-axis.

Combining inequality (13) and equation (18) gives us the following inequality:

$$||Q(s)||_\infty < \frac{1}{||e^{-sT_d} - 1||_\infty} \tag{19}$$

The chosen filter is given by the following equation:

$$Q(s) = \frac{\omega_n^2}{(\tau s)^2 + a(\tau s) + \omega_n^2} \tag{20}$$

where ω_n represents the bandwidth of the filter while a and τ are constants.
Choosing ω_n to be 50 rad/s, a to be 1000 and τ to be 0.4, the selected bandwidth ω_n limits the Q-filter's gain such that its magnitude, $|Q(j\omega)|$, never intersects or exceeds the magnitude of the inverse uncertainty bound, $|\Delta^{-1}(j\omega)|$, to satisfy the inequality (13). Since the uncertainty $\Delta(j\omega)$ dominated by unmodeled high-frequency dynamics and time delays grows with frequency, the stability limit $|\Delta^{-1}(j\omega)|$ drops significantly at higher frequencies. creating a stability limit which decreases the bandwidth when subjected to higher uncertainties, so according to the small gain theorem the robust stability of the system should be guaranteed[23]. The bode plots of $Q(s)$ and $\Delta^{-1}(s)$ are shown for X, Y and Z in figures 6, 7 and 8.

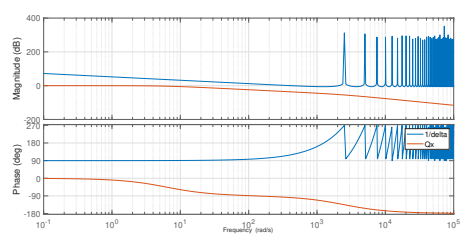

Fig. 6. Bode diagram of filter Q_x and uncertainty Δ_x^{-1}

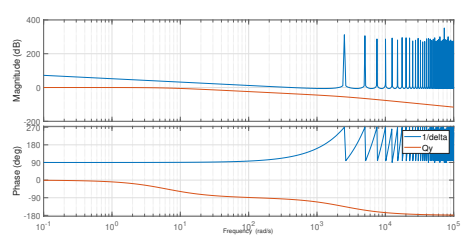

Fig. 7. Bode diagram of filter Q_y and uncertainty Δ_y^{-1}

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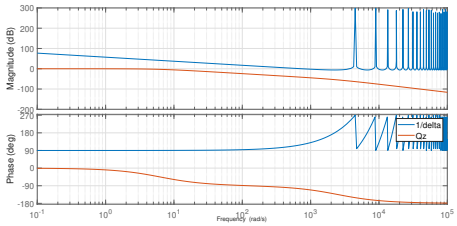

Fig. 8. Bode diagram of filter Q_z and uncertainty Δ_z^{-1}

TABLE I
SYSTEM CONSTANTS AND THEIR VALUES

Constant	Value	Constant	Value
A_x	0.25	l	0.225 m
A_y	0.25	m	0.468 kg
A_z	0.25	σ_u	2.12 m/s
I_{xx}	0.004856 m^4	σ_v	2.12 m/s
I_{yy}	0.004856 m^4	σ_w	1.4 m/s
I_{zz}	0.008801 m^4	L_u	200 m
b	1.14×10^{-7}	L_v	200 m
g	9.81 m/s^2	L_w	50 m
k	2.98×10^{-6}	V_a	0.5 m/s

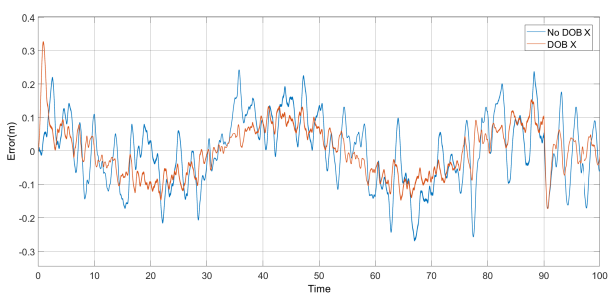

Fig. 10. Simulation results: Position error for the X direction

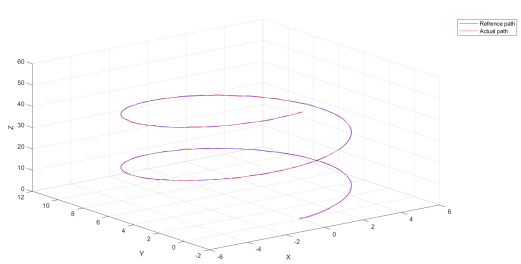

Fig. 9. Simulation results: UAV path

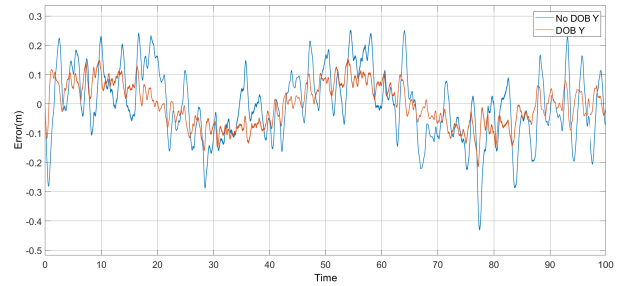

Fig. 11. Simulation results: Position error for the Y direction

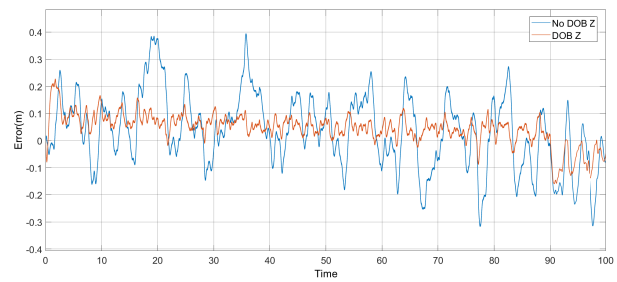

Fig. 12. Simulation results: Position error for the Z direction

V. SIMULATION RESULTS

Simulations were conducted to compare the performance of a quadrotor equipped with a disturbance observer (DOB)-based controller against that of a conventional cascaded PID controller. Disturbances were modeled using the Dryden wind turbulence model. The constants employed in the quadrotor model are listed in Table I. The PID values were chosen with the help of the MATLAB PID Tuner App to optimal values for rise time, settling time and overshoot. For the outer loop, the chosen values for the PID controllers were 4, 0.04, 2.4 for X and Y, and 3, 0.04, 2 for Z. For the inner loop, the chosen values for the PD controllers were 1 and 4.5. The simulated trajectory is shown in figure 9. Tracking errors for both controllers in the X, Y, and Z axes are compared in figures 10, 11, and 12, respectively.

The performance of the cascaded PID and DOB-based controllers was evaluated using RMSE and standard deviation metrics along the X, Y, and Z axes. The cascaded PID controller exhibited RMSE values of 10.2, 11.9, and 14 cm and standard deviations of 10.2, 11.9, and 13.4 cm, in the X, Y, and Z directions, respectively. In comparison, the DOB-based controller achieved lower RMSE values of 7.5, 7.3, and 7.2 cm and standard deviations of 7.5, 7.2, and 5.4 cm. Overall, the DOB-based controller reduced RMSE by 26.47%, 38.66%, and 48.57% in the X, Y, and Z positions, respectively. Corresponding standard deviation reductions of 26.47%, 39.5%, and 59.7% were observed.

VI. CONCLUSION

In this study, a disturbance observer–based control scheme was developed for a quadrotor UAV, and its robust stability was verified using the small-gain theorem. The proposed controller

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was evaluated against a conventional cascaded PID controller through simulation scenarios in which the UAV was required to track reference trajectories under continuous Dryden wind disturbances. The results demonstrated that the disturbance observer–augmented controller achieved superior stabilization and trajectory tracking performance compared to the baseline PID framework. Notably, the addition of the disturbance observer significantly enhanced the effectiveness of the otherwise simple cascaded PID architecture, highlighting its potential as a practical and robust solution for UAV control.

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VII. Appendix

The two matrices J and C are shown below where J represents the inertia matrix while C represents centripetal matrix:

$$J = \begin{bmatrix} I_{xx} & 0 & -I_{xx}S\theta \\ 0 & I_{yy}C^2\phi + I_{zz}S^2\phi & (I_{yy} - I_{zz})C\phi S\phi C\theta \\ -I_{xx}S\theta & (I_{yy} - I_{zz})C\phi S\phi C\theta & J_{33} \end{bmatrix} \tag{21}$$

$$J_{33} = I_{xx}S^2\theta + I_{yy}S^2\phi^2\theta + I_{zz}C^2\phi C^2\theta \tag{22}$$

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \tag{23}$$

$$C_{11} = 0 \tag{24}$$

$$\begin{aligned} C_{12} = &(I_{yy} - I_{zz})(\dot{\theta}C\phi S\phi + \dot{\psi}S^2\phi C\theta) \\ &+(I_{zz} - I_{yy})\dot{\psi}C^2\phi C\theta - I_{xx}\dot{\psi}C\theta \end{aligned} \tag{25}$$

$$C_{13} = (I_{zz} - I_{yy})\dot{\psi}C\phi S\phi C^2\theta \tag{26}$$

$$\begin{aligned} C_{21} = &(I_{zz} - I_{yy})(\dot{\theta}C\phi S\phi + \dot{\psi}S\phi C\theta) \\ &+(I_{yy} - I_{zz})\dot{\psi}C^2\phi C\theta + I_{xx}\dot{\psi}C\theta \end{aligned} \tag{27}$$

$$C_{22} = (I_{zz} - I_{yy})\dot{\phi}C\phi S\phi \tag{28}$$

$$\begin{aligned} C_{23} = &- I_{xx}\dot{\psi}S\theta C\theta + I_{yy}\dot{\psi}S^2\phi S\theta C\theta \\ &+ I_{zz}\dot{\psi}C^2\phi S\theta C\theta \end{aligned} \tag{29}$$

$$C_{31} = (I_{yy} - I_{zz})\dot{\psi}C^2\theta S\phi C\phi - I_{xx}\dot{\theta}C\theta \tag{30}$$

$$\begin{aligned} C_{32} = &(I_{zz} - I_{yy})(\dot{\theta}C\phi S\phi S\theta + \dot{\phi}S^2\phi C\theta) \\ &+(I_{yy} - I_{zz})\dot{\phi}C^2\phi C\theta + I_{xx}\dot{\psi}S\theta C\theta \\ &-I_{yy}\dot{\psi}S^2\phi S\theta C\theta - I_{zz}\dot{\psi}C^2\phi S\theta C\theta \end{aligned} \tag{31}$$

$$\begin{aligned} C_{33} =&(I_{yy} - I_{zz})\dot{\phi}C\phi S\phi C^2\theta - I_{yy}\dot{\theta}S^2\phi C\theta S\theta \\ &-I_{zz}\dot{\theta}C^2\phi C\theta S\theta + I_{xx}\dot{\theta}C\theta S\theta \end{aligned} \tag{32}$$

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